<table>
<tr><td>Marwadi University
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Faculty of Engineering &Technology
Department of Information and Communication Technology</td></tr>
<tr><td>Subject: Digital Signal and Image Processing (01CT1513)</td><td colspan="2">AIM: Simulate linear convolution and circular convolution on discrete time signals.</td></tr>
<tr><td>Experiment No: 3</td><td>Date:10-08-2026</td><td>Enrollment No:92400133120</td></tr>
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AIM: Simulate cross correlation and autocorrelation on discrete time signals.

Theory:

Cross-correlation and autocorrelation are mathematical operations used to measure the similarity or correlation between two signals. They are widely used in various applications, such as signal processing, image processing, and pattern recognition.

Cross-correlation measures the similarity between two signals at different time shifts. It computes the dot product of one signal with a time-shifted version of the other signal. The resulting cross-correlation signal indicates the similarity between the two signals at different time lags.

Autocorrelation, on the other hand, measures the similarity of a signal with a time-shifted version of itself. It computes the cross-correlation of a signal with itself. The autocorrelation signal shows how the signal is correlated with itself at different time lags.

Flowchart:

The flowchart for the program will consist of the following steps:

Define two discrete-time signals (signal 1 and signal 2) or use the same signal for autocorrelation.

Compute the cross-correlation of the two signals or the autocorrelation of a single signal. Plot and display the cross-correlation or autocorrelation signal.

Save the cross-correlation or autocorrelation signal (optional).

Now, let's see the Python program that simulates cross-correlation and autocorrelation on discrete-time signals:

Explanation:

We import the required modules, including numpy for array operations and matplotlib.pyplot for plotting.

The cross_correlation function computes the cross-correlation between two signals using numpy.correlate with the mode set to 'full'.

The autocorrelation function computes the autocorrelation of a single signal by calling the cross_correlation function with the same signal as both input arguments. In the main part of the program, we define two discrete-time signals (signal1 and signal2) or use the same signal for autocorrelation.

We compute the cross-correlation by calling the cross_correlation function with signal1 and signal2.

We compute the autocorrelation by calling the autocorrelation function with signal1. We plot and display the cross-correlation and autocorrelation signals using matplotlib.pyplot.stem.

The cross-correlation and autocorrelation signals can be optionally saved to files using numpy.savetxt.

You can modify the signal1 and signal2 variables to use different discrete-time signals for cross-correlation. Alternatively, you can use the same signal for autocorrelation by assigning signal1 to signal2.

Uncomment the np.savetxt lines if you want to save the cross-correlation or autocorrelation signals to text files.

Make sure you have numpy and matplotlib installed (pip install numpy matplotlib) before running this program.

Code:

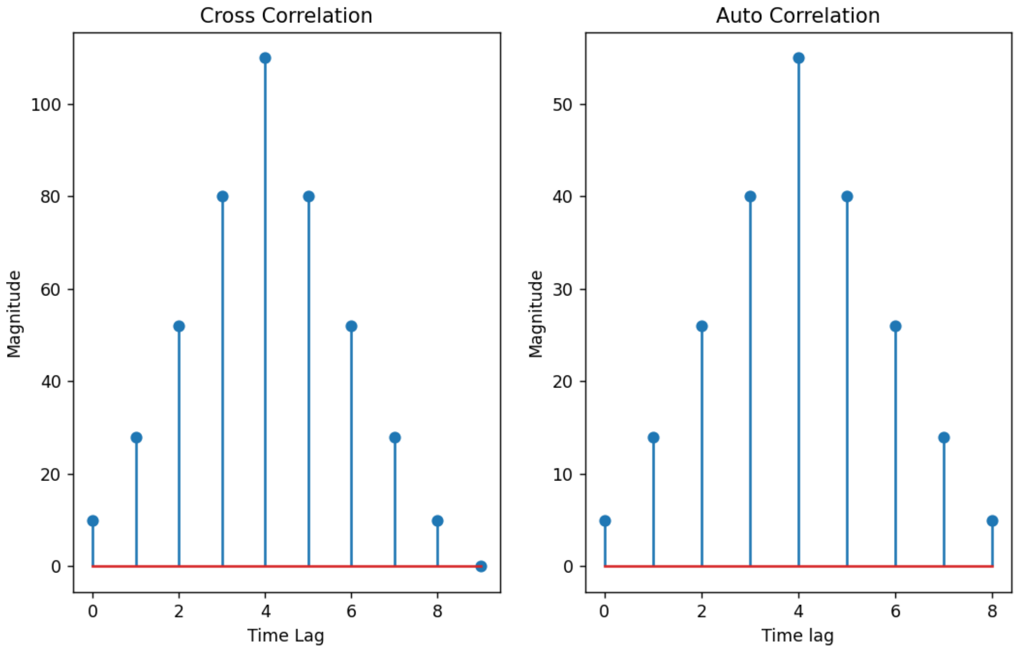
```
import numpy as np
import matplotlib.pyplot as plt

def cross_correlation(sig1,sig2):
    cross_corre=np.correlate(sig1,sig2,mode="full")
    return cross_corre
def auto_correlation(sig1):
    auto_corre=np.correlate(sig1,sig1,mode="full")
    return auto_corre
sig1=np.array([1,2,3,4,5])
sig2=np.array([0,2,4,6,8,10])

#cross correlation
cross=cross_correlation(sig1,sig2)
#auto correlation
auto=auto_correlation(sig1)
#plot both
plt.figure(figsize=(10,6))
plt.subplot(1,2,1)
plt.stem(cross)
plt.xlabel("Time Lag")
plt.ylabel("Magnitude")
plt.title("Cross Correlation")
```

```
plt.subplot(1,2,2)
plt.stem(auto)
plt.xlabel("Time lag")
plt.ylabel("Magnitude")
plt.title("Auto Correlation")
plt.show()
```

Output:



Conclusion:
In cross correlation we multiply two different discrete time signals and in case of auto correlation we multiply signal with it self. The program demonstrates both of these correlations. Here we have made two function auto_correlation and cross_correlation. These functions return the plot for cross correlated signal and auto correlated signal respectively. Next we have also defined two discrete time signals using numpy library as array. The above graph gives us the respective signals

Github link: